

Innovation, signaling, and virtual try-on: aligning immersive tools with a luxury brand

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471

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Abstract

Purpose – This paper aims to examine how Virtual Try-On (VTO) technologies shape consumers' attitudes and purchase intentions across brand types, by testing the mediating role of attitude toward VTO and the moderating role of brand type (luxury vs non-luxury) within the Technology Acceptance Model (TAM) and its experiential extension (e-TAM).

Design/methodology/approach – In this paper we develop a conceptual model grounded in e-TAM and test it through two experimental studies in the clothing and footwear categories. Analyses assess (1) the effects of e-TAM variables on attitudes and purchase intentions via mediation by evaluations of VTO and (2) the boundary condition of brand type.

Findings – The findings of this paper show that classic e-TAM variables significantly predict attitudes and purchase intentions when their effects operate through favorable evaluations of VTO. Brand type moderates this process: the e-TAM-to-attitude relationship is stronger for non-luxury than for luxury brands.

Originality/value – The paper presents integrate technology acceptance and brand signaling perspectives to contextualize immersive retail technologies by brand positioning. From an innovation management standpoint, it (1) establishes a contingency view of digital innovation payoff, showing that VTO's ROI is boundary-conditioned by brand type, and (2) offers innovation portfolio guidance, suggesting broad, faster rollouts for non-luxury brands and more selective, design-intensive pilots for luxury contexts. Offer actionable guidance for retailers on when and how to deploy VTO tools effectively across luxury and non-luxury settings.

Keywords Virtual try-on, Luxury brands, Experiential technology acceptance model, Signaling theory, Augmented reality, Consumer behavior

Paper type Research article

1. Introduction

Digital technologies have transformed how consumers interact with brands, particularly in online retail contexts where immersive tools like Virtual Try-On (VTO) are increasingly adopted. By allowing consumers to visualize products through augmented-reality (AR) interfaces, VTO acts as a “service augmentation” mechanism that it compensates for the lack of physical inspection in e-commerce (Hilken *et al.*, 2017), thereby enhancing perceived control, informativeness and vividness. Prior research consistently shows that VTO technologies can improve consumers' attitudes and related intentions (Yim *et al.*, 2017; Smink *et al.*, 2019; Holdack *et al.*, 2022), by activating both utilitarian evaluations (e.g. usefulness and ease of use) and hedonic responses (e.g. enjoyment and immersion), with recent evidence suggesting that the latter may outweigh the former (Costa *et al.*, 2025; Plotkina and Saurel, 2019). Although a growing consensus suggests that VTO technologies generate positive outcomes in online retail environment, much of the existing literature implicitly assumes that consumers' responses to VTO are driven by system-related evaluations (Jegadeesan *et al.*, 2025), thereby overlooking the heterogeneity of consumption context and psychological mechanisms that may differently shape attitudes towards VTO and purchase



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intentions (Batool and Mou, 2024; Sekri *et al.*, 2025). As a result, prior research has paid limited attention to the contextual factors that may condition how such experiential evaluations translate into favorable behavioral responses. This gap is particularly relevant with respect to the brand context, as different brand types may frame consumers' interaction with such immersive technologies, even if they do not directly alter the functional features of the VTO experiences (Sestino, 2024; Mishra *et al.*, 2025). Accordingly, this paper focuses on the role of brand type—specifically luxury versus non-luxury brand—in moderating consumers' attitudes toward VTO and their subsequent willingness to buy. Consistent with this, VTO tools are widely adopted by both luxury and non-luxury brands (Seegers, 2024), suggesting that such technologies are deployed across various brand domains with different symbolic meanings.

Prior work mainly treats the technology-attitude-intention link as context-invariant, leaving space for understanding whether the same experiential evaluations translate into different attitudes toward VTO depending on brand type. This gap aligns with emerging evidence in branding and user experience research showing that consumers' attitudes and motivations towards luxury brands are distinct from those in customers of non-luxury brands (e.g. Khelladi *et al.*, 2024). In non-luxury contexts, where brand value is mainly grounded in functional efficiency, informativeness, and convenience, immersive technologies such as VTO are likely to be evaluated in terms of their instrumental benefits (de Almeida, 2021; Costa *et al.*, 2025). In such contexts, positive evaluations of VTO are therefore more likely to translate into favorable attitudes toward technology. By contrast, in luxury contexts—where behavioral choices are shaped by experiential values, exclusivity, brand rarity, and the management of psychological distance (Ko *et al.*, 2019; Holmqvist *et al.*, 2020) - the evaluation of VTO is more strongly anchored in the symbolic meaning of the brand and in the perceived appropriateness of the technology within the shopping experience (Commuri, 2009; Shukla *et al.*, 2022). In this sense, VTO is mainly valued as a tool for preliminary engagement and digital information processing (Smink *et al.*, 2020), enabling consumers to explore products and interact with the brand prior to purchase while also elicit emotional reactions. Accordingly, although VTO can generate engagement and positive affective responses in luxury settings, its impact on consumers' behavior depends on the perceived congruence between VTO and the brand's positioning and experiential logic (Nann *et al.*, 2024). Against this backdrop, prior VTO research has rarely examined whether and how brand type directly shapes the formation of attitudes toward such technology, thereby leaving unclear whether experiential evaluations are interpreted differently across luxury and non-luxury contexts (Rauschnabel *et al.*, 2019; Sestino, 2024). Building on this gap, understanding consumers' responses to VTO requires shifting the focus to how experiential engagement is formed and unfolds during the shopping experience, rather than on functional performance alone. Building on the Technology Acceptance Model (TAM; Davis, 1989), the Experiential Technology Acceptance Model (e-TAM; Van der Heijden, 2000, 2004) extends the original framework by explicitly incorporating the experiential and hedonic dimensions of technology use. According to e-TAM, consumers' attitudes towards technology emerge from perceived usefulness (PU), perceived ease of use (PEOU), and perceived enjoyment (PE) which capture cognitive utility, interactional effort, and intrinsic hedonic value (van der Heijden, 2003, 2004). Although prior research generally finds that these core e-TAM dimensions are positively associated with purchase-related intentions (e.g. Nguyen *et al.*, 2025; Guo and Zhang, 2024), both the strength and the structure of this relationship may not be uniform across brand domains, based on how such experiential beliefs are interpreted within the broader technology-attitude-behavior chain (Yim *et al.*, 2017; Costa *et al.*, 2025). With such immersive technologies like VTO, these belief components tend to co-occur and jointly shape consumers' experiential evaluation of the interaction (Smink *et al.*, 2019; Holdack *et al.*, 2022). Consistent with research on hedonic and immersive systems (Van der Heijden, 2004; Lowry *et al.*, 2012), this dynamic allows us to conceptualize experiential technology acceptance as a higher-order construct reflected by perceived usefulness, perceived ease of use, and perceived enjoyment.

In such contexts, these beliefs collectively form an integrated appraisal (Moon and Kim, 2001; Venkatesh, 2000) that informs consumers' attitudes toward the technology, which serve as the key mechanism, linking experiential beliefs to purchase intentions (Davis, 1989; van der Heijden, 2003, 2004). This implies that willingness to buy should be understood as a downstream outcome of this technology-attitude chain, rather than as a direct or automatic consequence of experiential VTO acceptance.

Therefore, understanding how consumers perceive and respond to VTO depending on brand context is essential for clarifying when and how such immersive technologies translate into favorable behavioral outcomes. Accordingly, this study addresses the following research question: *How does the type of brand (luxury vs non-luxury) influence the relationship between experiential technology acceptance and consumer purchase intentions in a VTO context?*

To address this research question, we revisit the e-TAM model by integrating insights from signaling theory (Spence, 1973) and costly signaling theory (McAndrew, 2021). This integration allows us to theoretically define brand type as a boundary condition of the technology-attitude-purchase intention chain, clarifying how and when experiential technology acceptance translates into willingness to buy across different brand domains.

We test our conceptual framework across two experimental studies manipulating brand type (luxury vs non-luxury) in immersive VTO settings. Our findings provide both theoretical and practical contributions. First, it extends e-TAM, by identifying brand type as a boundary condition of the technology-attitude-behavior chain. Second, it bridges signaling theory with technology acceptance research, demonstrating how symbolic brand characteristics condition consumers' responses to immersive digital tools. Third, it advances understanding of VTO effectiveness by showing when immersive technologies reinforce, rather than weaken, brand-related evaluations. Finally, from a managerial point of view, our findings offer guidance for brand seeking to align immersive technologies with their positioning strategies.

The remainder of the paper is structured as follows. First, we develop our conceptual framework by reviewing the relevant literature and deriving our hypotheses. Next, we describe the research designs and report the results of two experimental studies design to test our predictions. Finally, we discuss the theoretical and managerial implications of our findings for technology adoption and brand management in digital retail.

2. Theoretical background

2.1 VTO and experiential technology acceptance

VTO represents a prominent application of augmented reality in online retail, enabling consumers to visually inspect and evaluate products prior to purchase (Zarkada et al., 2025). By recreating aspects of physical inspection in a digital environment, VTO bridges the gap between digital and physical shopping experiences (Kim and Forsythe, 2008a; Opute et al., 2022), thereby reducing uncertainty (Patnaik, 2024) and enhancing consumers' perceived control, confidence in product fit and appearance (Zhang et al., 2019; Kowalczyk et al., 2021). At the same time, its interactive nature stimulates positive affective responses, through entertainment and engagement (Zhang et al., 2019; Bialkova and Barr, 2022). Because of this dual nature, VTO delivers both utilitarian and hedonic benefits. On the one hand, it improves access to product-related information (Kim and Forsythe, 2008a; Zhang et al., 2019) and facilitates decision-making (Watson et al., 2020). On the other hand, it provides an immersive experience that increases intrinsic motivations and satisfaction (Başgömez and Yaman, 2022), positively influencing purchase intention (Merle et al., 2012; Zhang et al., 2019; Kim and Forsythe, 2007).

Given this combination of functional and experiential value, VTO is commonly conceptualized as an experiential digital system (Kim and Forsythe, 2008a). Consistent with this view and within both TAM and e-TAM frameworks (Kim and Forsythe, 2008a, b; Chen et al., 2024), attitude toward VTO represents the main evaluative response linking experiential beliefs to purchase intentions. In line with attitude theory, it refers to consumers'

overall favorable or unfavorable evaluation of using the VTO interface as a shopping mode (Eagly and Chaiken, 1993). Importantly, this construct is conceptually distinct from attitude toward the brand itself (Percy and Rossiter, 1997; Rossiter, 2014; Lavoye *et al.*, 2023), which reflects evaluations of the brand as a symbolic and functional object. Rather, attitude toward VTO captures the evaluation of the VTO-mediated interaction as a shopping means (Davis, 1989; van der Heijden, 2004). This implies that consumers may perceive VTO technology as useful and enjoyable while simultaneously questioning whether such an interaction mode is appropriate for engaging with a specific brand type.

2.2 Brand type as a symbolic context: luxury vs non luxury

Luxury and non-luxury brands differ not merely in price or quality, but in the type of value they convey and the psychological orientations they activate (Wiedmann *et al.*, 2009; Vigneron and Johnson, 2017). As such, in consumer behavior research (e.g. Tynan *et al.*, 2010; Park *et al.*, 2021) luxury and non-luxury brands underline distinct value configurations shaping how consumers interpret products, experiences and interaction modes (Kapferer and Laurent, 2016). Non-luxury brands are primarily evaluated through instrumental criteria such as functional performance, value-for-money, convenience, and risk-reduction (Kumagai and Nagasawa, 2019). In these contexts, consumers mainly focus on concrete and performance-related product attributes (Trope *et al.*, 2007) and brand relationships tend to be more transactional and utilitarian (Stokburger-Sauer and Teichmann, 2013). Luxury brands, by contrast, derive their value from symbolic and identity-relevant dimensions that support self-expression and status signaling (Stokburger-Sauer and Teichmann, 2013; Zhang and Cude, 2018). Their perceived value is anchored in exclusivity, heritage, craftsmanship cues, and controlled accessibility (Kapferer and Laurent, 2016). As a result, luxury brand tends to foster stronger brands attachment and identification processes (Moon and Sprott, 2016; Ko *et al.*, 2019), such that perceived luxury is positively associated with willingness to pay price premiums and engage in brand advocacy (Stokburger-Sauer and Teichmann, 2013; Shukla *et al.*, 2022).

Evaluations of luxury brands are therefore closely linked to the perception of authenticity and exclusivity. While authenticity refers to the extent to which a brand is seen as credible and consistent with its heritage and craftsmanship (Beverland, 2005; Morhart *et al.*, 2015), exclusivity relies on symbolic distance, supported by scarcity cues and selective distribution that enhance the consumers' perception of rarity (Trope and Liberman, 2010; Cowan and Kostyk, 2024). These dynamics align with signaling theory (Spence, 1973), which posits that individuals use observable attributes to communicate otherwise unobservable qualities, like wealth, taste, or social status. In luxury contexts, such brand ownership functions as a social signal aimed at influencing others' perceptions (Zhang and Cude, 2018). Crucially, the effectiveness of such signals depends on credibility. That is, in line with cost signaling theory (McAndrew, 2021), signals retain their value when they are costly or difficult to imitate. By maintaining scarcity and high price thresholds, luxury brands preserve the credibility of their symbolic signals (Kapferer and Laurent, 2016; Ko *et al.*, 2019). Because of this symbolic orientation, the way an experience is delivered becomes part of the brand meaning; when consumers pursue status affirmation and identity expression, they tend to prefer interaction modes that reinforce brand-self congruity at a symbolic level (Sirgy, 1982).

3. Conceptual model

Building on these theoretical foundations, our conceptual model examines how experiential technology acceptance translates into purchase intentions in a VTO context, and how this relationship may vary depending on brand type (luxury vs non-luxury; Rahman *et al.*, 2023). Because in VTO context perceived usefulness, ease of use and enjoyment are typically activated simultaneously during the interaction and jointly form an overall experiential

appraisal, we model experiential technology acceptance as a higher-order construct reflected by the three constructs, focusing on their combined evaluation rather than on separate effects (Kim *et al.*, 2007; Youn and Lee, 2019). Indeed, when consumers perceive VTO as useful, easy to use, and enjoyable, they experience greater confidence in product fit, which increases willingness to buy (Kim and Forsythe, 2008a; Smink *et al.*, 2019). Recent research (Kowalczyk and Hof, 2025) show that a simultaneous positive evaluation of usefulness, ease to use, and enjoyment is associated with higher value-in-use and stronger purchase intention. Accordingly, we formulate the following hypothesis:

- H1.* Overall e-TAM evaluation of VTO positively relates to consumers' willingness to buy.

Building on this value-based operationalization of overall experiential technology acceptance, previous literature suggests that experiential technology acceptance influences consumers' attitude toward relying on a digital interface into their shopping experiences (e.g. Lee *et al.*, 2006). When VTO is perceived as useful, easy to use, and enjoyable, consumers are more likely to evaluate the interface as a desirable and appropriate way to accomplish shopping tasks (Mariani *et al.*, 2023; Lee *et al.*, 2025), resulting in more favorable attitudes toward the technology. Formally, this led to the following hypothesis:

- H2.* Overall e-TAM evaluation is positively related to attitudes toward VTO technology.

Prior research consistently identifies attitude as the most proximal predictor of purchase intention (e.g. Ajzen, 1991; Chen *et al.*, 2024). Evidence further suggests that, even when cognitive and affective beliefs are simultaneously activated, once consumers have formed an overall evaluation relative to technology usage, this attitude becomes the enabler of purchase intention (e.g. Costa *et al.*, 2025; Nguyen *et al.*, 2025). Accordingly, we hypothesize:

- H3.* Attitude toward VTO is positively related to consumers' willingness to buy.

Taken together, these arguments suggest that overall experiential technology acceptance can affect willingness to buy both directly and indirectly through the mediating role of attitude toward VTO (e.g. Davis, 1989; van der Heijden, 2003). Because attitudinal judgments integrate cognitive and affective responses into an overall readiness to rely on the technology within the shopping process (Moon and Kim, 2001), we expect attitude to mediate the relationship between overall e-TAM and willingness to buy. Therefore, we propose:

- H4.* Attitude toward VTO mediates the relationship between overall e-TAM evaluation and consumers' willingness to buy.

The extent to which such evaluation translates into a favorable attitude toward the technology also depends on the symbolic architecture within which such innovations are embedded. Drawing on signaling theory (Spence, 1973) and costly signaling theory (McAndrew, 2021), we propose that luxury brands, which derive value from exclusivity and status signaling, may see diminished consumer responses to VTO if the technology is perceived as overly democratizing or incongruent with brand prestige. In contrast, non-luxury brands may benefit more directly from the practical and hedonic advantages of VTO, as consumers seek utility and personalization. While signaling theory has been used extensively to explain why consumers pay premiums for luxury as status cues, it has rarely been connected to technology acceptance models in immersive retail. Yet, luxury value depends on signals that are credible because they are difficult to mimic (costly signaling) and because brands manage information, access, and scarcity strategically (Spence, 1973; Nelissen and Meijers, 2011; Han *et al.*, 2010). This implies a specific tension for VTO: by making product experience more accessible and replicable, VTO may reduce perceived exclusivity/brand essence, thereby weakening the usual positive translation from e-TAM beliefs into attitudes and willingness to buy in luxury contexts (Nann *et al.*, 2024). In this regard, brand type (luxury vs non-luxury) represents a boundary condition shaping how new immersive technologies are interpreted. In non-luxury

contexts, since experiential beliefs are likely to be evaluated in terms of direct contribution to efficiency and confidence, they are easier to translate into favorable attitude toward VTO (Kumagai and Nagasawa, 2019). By contrast, because luxury brands derive their value from authenticity, exclusivity and controlled accessibility (Wiedmann *et al.*, 2009), the deployment of such immersive technologies may be perceived as increasing accessibility and standardization. Such democratization can weaken consumers' exclusivity perception (Commuri, 2009; Shukla *et al.*, 2022), thereby attenuating the extent to which positive experiential evaluations of VTO translate into favorable attitudes toward the technology itself (Rosendo-Rios and Shukla, 2023). Accordingly, we posit the following hypothesis:

- H5. The positive relationship between overall e-TAM evaluation and attitudes toward VTO is expected to be weaker for luxury brands than for non-luxury brands.

The conceptual model illustrating these relationships is presented in Figure 1.

4. Methodology

4.1 Pre-test of the luxury versus non-luxury brand manipulation

To operationalize brand type using real fashion brands, we first conducted a pre-test. This step was important because real brands may vary not only in luxury status but also in familiarity, target-audience fit, price positioning, and other brand-specific associations. The pre-test was designed to assess whether the selected brands could be meaningfully distinguished on perceived luxury and to provide evidence that the luxury versus non-luxury manipulation was not driven solely by idiosyncratic brand meanings. Therefore, we assessed perceived luxury using a multi-item scale and triangulated the results with a brand-positioning measure and a direct single-item luxury check. Perceived luxury was measured using five items ($\alpha = 0.947$) assessing whether the brand was perceived as luxurious, prestigious, high-class, exclusive, and premium. This multi-item assessment is consistent with prior luxury-brand research (e.g. Vigneron and Johnson, 2017; Christodoulides *et al.*, 2009; Hagtvedt and Patrick, 2009) that conceptualizes perceived luxury as a multidimensional construct involving prestige, exclusivity, high-end positioning, and premium perceptions. In addition, brand positioning was measured using three semantic differential items adapted from Hagtvedt and Patrick (2016): economical–expensive, low-end–high-end, and affordable–luxurious ($\alpha = 0.872$). Finally, a direct single-item manipulation checks asked participants to rate the brand on a continuum from non-luxury brand to luxury brand.

Data was collected through structured online questionnaires administered via Qualtrics. The survey link was disseminated through social media platforms (WhatsApp, Instagram, and Facebook) and email invitations by one of the authors. This approach is consistent with prior online survey research that used digital channels for respondent recruitment (e.g. Abbate *et al.*,

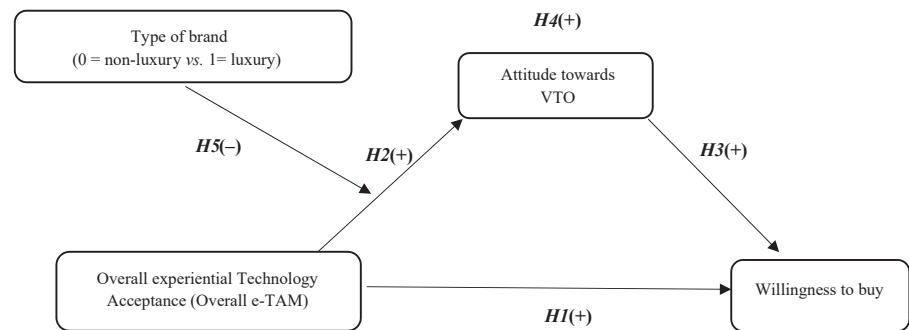


Figure 1. Conceptual model. Source: Authors' own work

2026). Participants ($N = 127$, 57.6% female, $M_{age} = 28.65$, $SD_{age} = 9.60$) were recruited in Italy through a convenience sampling approach combined with a snowball technique (Sadler, 2010; Parker *et al.*, 2019) and were randomly assigned to one of three between-subjects brand conditions: Zara®, Gucci®, or Valentino®. Zara® was selected to represent the non-luxury fashion brand condition, whereas Gucci® and Valentino® were selected to represent the luxury fashion brand condition. The inclusion of two luxury brands was intended to reduce the risk that the effect was driven by idiosyncratic associations with a single luxury brand. At the same time, the pre-test explicitly assessed whether the two luxury brands were perceived similarly on luxury-related dimensions and whether both were clearly differentiated from Zara®.

The first one-way ANOVA revealed a significant effect of brand condition on perceived luxury, $F(2, 124) = 82.96$, $p < 0.001$. As expected, Zara® was perceived as substantially less luxurious ($M = 2.77$, $SD = 1.25$) than Gucci® ($M = 5.52$, $SD = 1.15$) and Valentino® ($M = 5.57$, $SD = 1.08$). Planned contrasts confirmed that Zara® differed significantly from the two luxury brands combined, $t(124) = 12.88$, $p < 0.001$, whereas Gucci® and Valentino® did not differ from each other, $t(124) = -0.19$, $p = 0.853$. Post-hoc comparisons further showed that Zara® was rated significantly lower than both Gucci® and Valentino®, while Gucci® and Valentino® belonged to the same homogeneous subset [1].

The same pattern emerged for the brand-positioning scale. The second one-way ANOVA showed a significant effect of brand condition, $F(2, 124) = 70.06$, $p < 0.001$. Zara® received lower positioning scores ($M = 3.57$, $SD = 0.96$) than Gucci® ($M = 6.01$, $SD = 0.82$) and Valentino® ($M = 5.68$, $SD = 1.32$). Because Levene's test was significant, Games-Howell post-hoc comparisons were inspected. Results confirmed that Zara® was rated significantly lower than both Gucci® and Valentino®, whereas Gucci® and Valentino® did not differ significantly from each other. The planned contrast comparing Zara® with the two luxury brands combined was significant, $t(98.95) = 12.14$, $p < 0.001$, while the contrast between Gucci® and Valentino® was not significant, $t(64.92) = 1.36$, $p = 0.177$.

The direct single-item manipulation check also confirmed the expected pattern. The third one-way ANOVA revealed a significant effect of brand condition, $F(2, 124) = 82.37$, $p < 0.001$. Zara® was perceived as less luxury-oriented ($M = 3.20$, $SD = 1.36$) than Gucci® ($M = 6.29$, $SD = 0.90$) and Valentino® ($M = 5.75$, $SD = 1.30$). Since Levene's test was significant, Games-Howell post-hoc comparisons were used. Results showed that Zara® differed significantly from both Gucci® and Valentino®. The comparison between Gucci® and Valentino® was not significant under Games-Howell, $p = 0.081$.

Overall, the pre-test provides evidence that the selected brands differ strongly and consistently on perceived luxury. Importantly the same pattern was observed across a reliable multi-item perceived luxury scale, a reliable brand-positioning scale, and a direct luxury manipulation check. Moreover, Gucci® and Valentino® were generally perceived as similarly luxury-oriented, suggesting that the luxury condition was not driven by one specific luxury brand. These results support the validity of the luxury *versus* non-luxury brand manipulation used in the main studies.

4.2 Experimental design approach

To test the proposed hypotheses, two between-subjects' experimental studies in a VTO online shopping scenario. Both studies manipulated brand type (non-luxury vs luxury) and adopted a comparable approach and measurement strategy to facilitate replication across product category. Study 1 focused on the clothing category, while Study 2 replicated the design in the footwear category. Across studies, the luxury condition was operationalized using different brands: Study 1 - Valentino® and Study 2 - Gucci®, while Zara® was used as the non-luxury benchmark in both studies. This reduces potential single-brand idiosyncrasies and strengthens inference regarding brand type as a positioning dimension rather than a brand-specific effect. Data was collected through structured online questionnaires administered via Qualtrics. The

online survey was chosen to ensure contextual alignment with the digital environment in which VTO technology is typically experienced (Kim and Forsythe, 2008a). Moreover, online data collection facilitates access to digitally engaged individuals who are more likely to be familiar with such immersive technologies. The Qualtrics platform enabled automated exclusion of incomplete responses and item-order randomization, that allow to reduce potential common method bias (Podsakoff et al., 2003), thereby improving data reliability (Braun et al., 2021).

Participants were recruited in Italy through a convenience sampling approach combined with a snowball technique (Sadler, 2010; Parker et al., 2019) and were randomly assigned to one of the experimental conditions. This non-probability method is widely adopted in behavioral and technology adoption research (Hultsch et al., 2002; Zhao, 2021) to efficiently reach consumers willing to participate in the study (Landers and Behrend, 2015; Scholtz, 2021) and examine theory-driven causal relationships. Although non-probability sampling may limit statistical generalizability of findings (Staetsky, 2019), this strategy ensures adequate coverage of respondents consistent with the research context (Etikan et al., 2016). The survey link was disseminated through social media platforms (WhatsApp, Instagram, and Facebook) and email invitations. This approach is consistent with prior online survey research that used digital channels for respondent recruitment (e.g. Abbate et al., 2026) and is particularly appropriate in the present study because Virtual Try-On (VTO) technologies are typically experienced in digital retail environments. Accordingly, recruiting participants through online communication channels ensured coherence between the data collection procedure and the technology-mediated shopping context under investigation.

Prior to participating, respondents were informed that the study aimed to assess perceptions related to the use of new digital technologies in online shopping experiences, without disclosing the specific hypotheses. Participation was voluntary and anonymous, and electronic informed consent was obtained before the start of the survey; after exposure to the stimulus, they completed the questionnaire.

4.3 Study 1: the effect of the experiential technology acceptance model of virtual try-on tools on consumers' willingness to buy and the brand type as a boundary condition

4.3.1 Participants and design. Data was collected in April 2025. A total of 250 responses were obtained. The survey link was disseminated through the network of another author via email invitations and social media (e.g. Abbate et al., 2026), and further circulated through a virtual snowball sampling approach (Sadler, 2010). Following well-established screening procedures in online behavioral research (e.g. Landers and Behrend, 2015), incomplete questionnaires and responses with unrealistically short completion times were excluded. The final sample includes 234 responses, corresponding to a response (usable) rate of 93.6% (55.6% female, $M_{age} = 38.38$, $SD_{age} = 14.78$), randomly assigned to one of the two conditions in a 2 (Brand type: non-luxury vs luxury) between-subjects design.

A priori power analysis using G*Power 3.1 (Faul et al., 2009) indicated that a minimum of 128 participants (64 per condition) was required to detect a medium effect size (Cohen's $d = 0.50$) with 80% power at $\alpha = 0.05$. The final sample size exceeded this threshold, indicating adequate statistical power to detect the hypothesized effects.

The manipulation of the brand type was operationalized presenting participants with either a non-luxury brand (Zara®) or a luxury brand (Valentino®). Each condition included a standardized scenario describing an online shopping experience involving the use of a VTO technology [2]. Participants were explicitly instructed to imagine themselves browsing the brand's website and using the VTO to virtually try on clothing items. They were asked to evaluate the experience under the assumption that the product matched their taste, and they had the financial means to purchase it. To enhance scenario realism, the description was accompanied by visual stimuli depicting a smartphone illustrating the VTO functionality.

After exposure to the scenario, participants completed the questionnaire. All constructs were measured using a seven-point Likert scale (1 = strongly disagree; 7 = strongly agree) adopting validated scales from previous studies. Perceived usefulness (PU) was measured with a four-items scale, perceived ease of use (PEOU) through a three-items scale and perceived enjoyment through a four-items scale. All the scales derive from Kim and Forsythe (2008b). Attitude towards VTO was measured through a four items scale by Chidambaram *et al.* (2024). Willingness to buy was measured using three items scale by Dodds *et al.* (1991–1 = very low; 7 = very high). All scales show satisfactory internal consistency (see Appendix for details). Consistent with established practices in scale-based research, item scores were averaged to compute composite indices for each construct (Diamantopoulos *et al.*, 2012). Then, in line with prior work on experiential and hedonic technology acceptance – where PU, PEOU and PE are interrelated beliefs jointly shaping an overall experiential appraisal (Venkatesh, 2000; van der Heijden, 2000, 2004) – we operationalized overall e-TAM as the mean of these constructs. This composite index exhibits satisfactory internal consistency ($\alpha = 0.81$).

Moreover, to assess the effectiveness of the experimental manipulation, participants completed a brand-type manipulation check on a “single-item luxury perception scale (i.e. ‘How would you rate the brand shown in this survey? 1 = non-luxury brand, 7 = luxury brand’)”. Finally, participants provided socio-demographic information (gender, age and education).

Furthermore, to mitigate Common Method Variance (CMV), Harman’s single-factor test was performed through an unrotated principal component analysis including all measurement items (18 items). The first factor accounted for 60.41% of the total variance ($N = 234$). Given that Harman’s test is a conservative diagnostic and may reflect substantive covariance among conceptually related constructs, we complemented this analysis with a marker-variable approach. Education was included as a theoretically unrelated marker. The marker variable showed negligible and non-significant correlations with the focal constructs (Education–Overall e-TAM: $r = -0.048$, $p = 0.470$; Education–ATT: $r = -0.047$, $p = 0.474$; Education–WTB: $r = -0.025$, $p = 0.702$). Moreover, partial correlations among the focal constructs controlling for education remained virtually unchanged and statistically significant (Overall e-TAM–ATT: $r = 0.803$, $p < 0.001$; Overall e-TAM–WTB: $r = 0.731$, $p < 0.001$; ATT–WTB: $r = 0.669$, $p < 0.001$). These results suggest that CMV does not pose a substantial threat to the validity of the findings.

4.3.2 Results and discussion. To verify the effectiveness of the brand-type manipulation, an independent samples *t*-test was conducted. Results show a statistically significant difference between the two conditions, $t(232) = -5.61$, $p < 0.001$, with participants in the non-luxury condition perceiving the brand as significantly less luxurious than those in the luxury condition (95% C.I. $[-1.64, -0.79]$). The observed effect size was large (Cohen’s $d = 0.74$, power = 0.99), confirming that the manipulation successfully included the perception brand type.

Hypotheses H1 - H4 were tested using PROCESS MACRO (Model 4; Hayes, 2017). Consistent with H1, overall e-TAM exerted a significant total effect on willingness to buy ($\beta = 0.64$, $t = 7.41$; $p < 0.000$), with a 95% bias-corrected confidence interval (BCCI) ranging from 0.47 (LLCI) to 0.81 (ULCI). The model explains 55.3% of the variance in willingness to buy ($R^2 = 0.553$; $F(2,231) = 143.16$; $p < 0.000$). Similarly, supporting H2, overall e-TAM significantly and positively influenced attitude toward VTO ($\beta = 0.85$; $t = 20.56$; $p < 0.000$), accounting for 64.6% of the variance ($R^2 = 0.646$; $F(1,232) = 422.58$; $p < 0.000$). In line with H3, attitude towards VTO significantly predicted willingness to buy ($\beta = 0.25$; $t = 3.10$; $p < 0.001$). Consistent with H4, the indirect effect of overall e-TAM on willingness to buy via attitude VTO was significant (indirect effect = 0.22), with a 95% bias-corrected bootstrap confidence interval based on 5,000 resamples of $[0.09; 0.36]$. To test H5, we estimated a moderated-mediation model using PROCESS MACRO (Model 7) in which brand type (non-luxury vs luxury) moderates the path from overall e-TAM to attitude toward VTO. The model

predicting willingness to buy accounted for 55.3% of the variance ($R^2 = 0.553$; $F(2,231) = 143.16$; $p < 0.001$). Results indicated that overall e-TAM positively predicted attitude toward VTO ($\beta = 0.95$, $p < 0.001$), and attitude in turn predicted willingness to buy ($\beta = 0.25$, $p < 0.001$). Importantly, even when including attitude in the model, overall e-TAM remained a significant predictor of willingness to buy ($\beta = 0.64$, $p < 0.001$), indicating partial mediation. Crucially, the interaction between overall e-TAM and brand type was negative and significant ($\beta = -0.17$, $p < 0.05$), thereby supporting H5. Specifically, this relationship between overall e-TAM and attitude towards VTO was stronger for the non-luxury brand than for the luxury brand ($\beta_{\text{non-luxuryBrand}} = 0.95$, $t = 15.22$, $p < 0.000$ vs $\beta_{\text{luxuryBrand}} = 0.78$, $t = 14.07$, $p < 0.000$). The indirect effects of overall e-TAM on willingness to buy via attitude towards VTO was statistically significant in both conditions, but weaker in the luxury condition ($\beta_{\text{non-luxuryBrand}} = 0.24$, $SE = 0.07$, $CI [0.10; 0.39]$ vs $\beta_{\text{luxuryBrand}} = 0.20$, $SE = 0.06$, $CI [0.08; 0.32]$). To further assess the conditional indirect effect, we examined the index of moderator mediation. It was negative and statistically significant ($\beta = -0.04$, $SE = 0.02$; 95% bias-corrected $CI [-0.10, -0.02]$), as the confidence interval did not include zero.

Overall, the results of Study 1 provided support for all proposed hypotheses and confirmed that greater acceptance of new technology (i.e. VTO) leads consumers to increase their willingness to buy the product. This relationship is explained by the fact that a greater acceptance of VTO technology positively affects the consumer's attitude toward new technology, which in turn, positively impacts the willingness to buy the product. In addition, our results provide evidence that the brand typology (non-luxury vs luxury) plays a moderating role in the relationship between acceptance of new technology and attitude towards new technology. The results suggest that this relationship is stronger in the case of non-luxury brand than in luxury ones.

To further assess the stability of the findings, we re-estimated the moderated-mediation model (using PROCESS MACRO–Model 7) including personal innovativeness toward technology, as well as socio-demographic controls (age, gender, education, and income). Personal innovativeness toward technology was measured with three items adapted from Kim and Forsythe (2008a). This construct captures respondents' general propensity to try and experiment with new technologies and therefore helps rule out the alternative explanation that the observed effects are merely driven by individual differences in technology orientation. The scale showed good internal consistency ($\alpha = 0.88$). The substantive pattern remained unchanged. In the first-stage model, overall e-TAM positively predicted attitude toward VTO ($\beta = 0.93$, $p < 0.001$), and the interaction between e-TAM and brand type remained negative and significant ($\beta = -0.18$, $p < 0.05$). The conditional effect of e-TAM on attitude toward VTO remained stronger for the non-luxury brand ($\beta = 0.93$, $p < 0.001$) than for the luxury brand ($\beta = 0.75$, $p < 0.001$). In the second-stage model, attitude toward VTO continued to predict willingness to buy positively ($\beta = 0.25$, $p < 0.01$). Moreover, the conditional indirect effects remained significant in both the non-luxury condition (effect = 0.24, 95% $CI [0.09, 0.38]$) and the luxury condition (effect = 0.19, 95% $CI [0.07, 0.31]$). Finally, the index of moderated mediation remained negative and significant (index = -0.05 , 95% $CI [-0.10, -0.003]$). Together, these findings suggest that the moderated-mediation effect is robust to controls for individual innovativeness and socio-demographic differences.

4.4 Study 2: the generalizability of the proposed effects on new product category

4.4.1 Participants and design. Study 2 was conducted to provide additional support for H1-H5 and to test the generalizability of the proposed effects across a different product category.

Data were collected in May 2025 through an online survey administrated via Qualtrics. The survey link was disseminated through the network of another author via email invitations and social media (e.g. Abbate *et al.*, 2026), and further circulated through a virtual snowball sampling approach (Sadler, 2010). Incomplete questionnaires and responses with unrealistically short competition times were excluded. Of the 191 collected questionnaires,

179 were retained for analysis (usable response rate = 93.7%), of which 52% female ($M_{age} = 40.08$, $SD_{age} = 14.59$). The sample size exceeds the minimum threshold (128 participants, 64 per condition) required by power analysis with G*Power 3.1 (Faul *et al.*, 2009). Participants were randomly assigned to one of two experimental conditions: the non-luxury brand (Zara®) or a luxury brand (Gucci®) [3]. All constructs were measured using the same validated scales adopted in Study 1. Internal consistency was satisfactory (PU $\alpha = 0.92$; PEOU $\alpha = 0.92$; PE $\alpha = 0.94$; ATT $\alpha = 0.86$; WTB $\alpha = 0.93$) - (see Appendix). As in Study 1, item scores were aggregated to compute composite indices. Overall e-TAM was operationalized as the mean of perceived usefulness, ease of use and enjoyment ($\alpha = 0.80$). As in Study 1, manipulation checks for brand type and socio-demographic variables (age, gender and education) were included at the end of the questionnaire.

Finally, common method variance was assessed using the same diagnostic strategy adopted in Study 1 (Harman's one-factor test and a marker-variable approach with education). Results converged in indicating that CMV does not substantially alter the estimated effects: Harman first factor = 59.27% of variance; marker correlations are trivial and non-significant (Education–Overall e-TAM: $r = -0.009$, $p = 0.904$; Education–ATT: $r = 0.001$, $p = 0.992$; Education–WTB: $r = 0.001$, $p = 0.989$), and partial correlations remained stable (Overall e-TAM–ATT $r = 0.789$, Overall e-TAM–WTB $r = 0.700$, ATT–WTB $r = 0.651$, all $p < 0.001$).

4.4.2 Results and discussion. The independent-samples *t*-test on brand type perception revealed a statistically significant difference between conditions, $t(177) = -4.44$, $p < 0.001$. Participants exposed to the non-luxury brand reported lower luxury perceptions ($M = 3.16$, $SD = 1.69$) than those exposed to the luxury brand condition ($M = 4.28$, $SD = 1.68$), with a 95% confidence interval for the mean difference of $[-1.61, -0.62]$. This difference corresponds to a medium-to-large effect (Cohen's $d = 0.66$) and was detected with high post hoc power ($\alpha = 0.05$; power = 0.993).

Results of a mediation analysis (Hayes, 2017 - PROCESS MACRO Model 4) yielded a significant total effect of overall e-TAM on willingness to buy ($\beta = 0.59$, $t = 5.77$; $p < 0.000$), 95% bias-corrected confidence interval (BCCI) $[0.39, 0.79]$. The model accounted for 51.5% of the variance in willingness to buy ($R^2 = 0.515$; $F(2, 176) = 93.45$; $p < 0.000$), supporting H1. Consistent with H₂, overall e-TAM also significantly and positively influenced attitude toward VTO ($\beta = 0.84$; $t = 17.10$; $p < 0.000$; $R^2 = 0.623$; $F(1, 177) = 292.66$; $p < 0.000$). Attitude toward VTO, in turn, was positively related to willingness to buy ($\beta = 0.29$; $t = 3.05$; $p = 0.0026$), supporting H3. In line with H₄, the indirect effect of overall e-TAM on willingness to buy through attitude toward VTO was significant (0.25) with a 95% bias-corrected bootstrap confidence interval based on 5,000 resamples of $[0.11; 0.40]$. In the moderated-mediation analysis (PROCESS MACRO–Model 7), willingness to buy accounted for 51.5% of the variance ($R^2 = 0.515$; $F(2, 176) = 93.45$; $p < 0.001$). Overall e-TAM positively influenced the attitude toward VTO technology ($\beta = 0.97$, $p < 0.001$), which in turns predicted willingness to buy ($\beta = 0.29$, $p = 0.0026$). Overall e-TAM retained a significant direct effect on willingness to buy ($\beta = 0.59$, $p < 0.001$). With respect to H5, the overall e-TAM * brand type interaction was negative and significant ($\beta = -0.22$, $p = 0.030$), indicating a stronger effect for the non-luxury brand compared to luxury brand ($\beta_{non-luxuryBrand} = 0.97$, $t = 12.79$, $p < 0.000$ vs $\beta_{luxuryBrand} = 0.75$, $t = 11.74$, $p < 0.000$). Also the indirect effect was significant in both conditions ($\beta_{non-luxuryBrand} = 0.28$, $SE = 0.08$, CI $[0.12; 0.46]$ vs $\beta_{luxuryBrand} = 0.22$, $SE = 0.07$, CI $[0.09; 0.35]$). Finally, to further verify whether the indirect effect differs across brand types, we examined the index of moderation-mediation. It was negative and significant ($\beta = -0.06$, $SE = 0.03$; 95% bias-corrected bootstrapped CI $[-0.14, -0.01]$).

Overall, the results of Study 2 provided further support for all proposed hypotheses by confirming the effect of the experiential technology acceptance model of virtual try-on tools is positively related to consumers' willingness to buy through a VTO attitude-based mechanism. Moreover, the findings of Study 2 allow for generalizing the predicted effect and mechanism

on a new product category, i.e. footwear. More importantly, our results provide further evidence that the brand type (non-luxury vs luxury) plays a moderating role in the relationship between acceptance of VTO technology and attitude towards VTO. The results suggest that this relationship is stronger in the case of non-luxury brand than in luxury ones also considering a different product category (i.e. footwear).

As in Study 1, we further re-estimated the moderated-mediation model (PROCESS MACRO–Model 7) by including personal innovativeness toward technology ($\alpha = 0.86$) as well as socio-demographic controls (age, gender, education, and income). Results remained substantively unchanged. In the first-stage equation, TAM positively predicted attitude toward VTO ($\beta = 0.97, p < 0.001$), and the interaction between TAM and brand type remained negative and significant ($\beta = -0.24, p < 0.05$), indicating that the positive effect of TAM on attitude toward VTO was weaker in the luxury-brand condition than in the non-luxury-brand condition. The conditional effects showed that the relationship was stronger for the non-luxury brand ($\beta = 0.97, p < 0.001$) than for the luxury brand ($\beta = 0.72, p < 0.001$). In the second-stage equation, attitude toward VTO remained a significant predictor of willingness to buy ($\beta = 0.30, p < 0.01$), and the direct effect of TAM on willingness to buy also remained significant ($\beta = 0.54, p < 0.001$). The conditional indirect effects were significant in both the non-luxury condition (effect = 0.30, 95% CI [0.13, 0.47]) and the luxury condition (effect = 0.22, 95% CI [0.10, 0.35]). Importantly, the index of moderated mediation remained negative and significant (index = -0.07 , 95% CI [$-0.16, -0.01$]). Taken together, these findings provide additional evidence that the focal pattern is robust to individual differences in technology innovativeness and basic socio-demographic characteristics.

5. General discussion

This research investigated how consumers' acceptance of Virtual Try-On (VTO) technologies is shaped by both experiential perceptions grounded in the Experiential Technology Acceptance Model (e-TAM) and the symbolic meaning conveyed by brand positioning. Building on signaling theory (Spence, 1973) and costly signaling theory, we tested a moderated mediation model across two experimental studies in the apparel (Study 1) and footwear (Study 2) sectors. Rather than treating the findings as a set of isolated statistical effects, the overarching picture that emerges is that VTO adoption is driven by a combination of (1) how useful, easy, and enjoyable the technology feels and (2) whether this experience "fits" what the brand is supposed to represent. The results consistently demonstrate that overall e-TAM – comprising perceived usefulness (PU), perceived ease of use (PEOU), and perceived enjoyment (PE) – significantly enhances consumers' willingness to buy (WTB), primarily through the mediating role of attitudes toward VTO technology. Importantly, this relationship is moderated by brand type (non-luxury vs luxury), with stronger effects observed in the context of non-luxury brand. Additionally, our findings imply a concrete roadmap for the further development of VTO and luxury product strategies. To align VTO with the signaling logic of luxury, access and interaction should be selectively scarce (such as invitation-only try-on sessions, capsule- or collection-specific filters that foreground craftsmanship narratives, and frequency caps that prevent overexposure), so that the experience itself functions as a status cue rather than a purely utilitarian tool. Moreover, algorithmic curation should optimize for self-brand fit and signaling congruence (not only click-through or conversion), with experimentation that explicitly tracks brand-equity safeguards (perceived exclusivity, brand-to-self distance) alongside standard KPIs (such as engagement, returns). Finally, omnichannel orchestration (e.g. virtual appointments with human stylists, digital certificates of authenticity linked to try-on artifacts, and limited-time immersive previews tied to boutique events) can reposition VTO technology from a transactional aid to a curated, craft-centered service that complements, rather than commodifies, luxury.

In terms of our hypotheses, Study 1 and Study 2 converge on a coherent mechanism. First, overall e-TAM was positively associated with consumers' willingness to buy (supporting H1),

indicating that favorable experiential appraisals of VTO translate into stronger purchase intentions. Second, overall e-TAM also fostered more positive attitudes toward VTO (supporting H2), suggesting that consumers' evaluations of usefulness, ease, and enjoyment are reflected in their overall disposition toward the technology. Third, attitudes toward VTO were positively related to willingness to buy (supporting H3), highlighting attitudes as a proximal driver of behavioral intention. Together, these links establish a consistent indirect pathway whereby overall e-TAM increases willingness to buy through attitudes (supporting H4). Finally, we found that the strength of the overall e-TAM → attitude link depends on brand type (supporting H5), meaning that the same technological experience can be interpreted differently depending on the symbolic frame provided by the brand.

These findings contribute to a contextualized understanding of technology acceptance in digital retail environments. Whereas traditional e-TAM applications emphasize system-related drivers as isolated antecedents of behavioral intention (Davis, 1989; Van der Heijden, 2000, 2004), our study shows that consumers' experiential evaluations of VTO are filtered through brand-related cognitive schemas. In other words, it seems that consumers do not simply ask "Is this technology good?"—they also ask "Does this technology make sense for this kind of brand?". In this sense, technological affordances are not interpreted in a vacuum but are meaningfully shaped by the symbolic associations consumers attach to the brand offering the experience.

In Study 1 (apparel), we found that the positive effect of e-TAM on attitudes toward VTO – and its indirect effect on WTB – was significantly stronger for a non-luxury brand (Zara®) than for a luxury brand (Valentino®). Study 2 replicated this moderated mediation pattern in a different product category (footwear), further confirming that the impact of e-TAM on consumer behavior is contingent on brand context. These results suggest that for non-luxury brands, which are typically associated with functionality, value-for-money, and practical decision-making (Kumagai and Nagasawa, 2019), VTO technology is perceived as congruent and valuable. In this context, consumers appear to appreciate the way VTO enhances personalization, reduces uncertainty, and offers an interactive shopping experience, thereby improving attitudes toward the tool and increasing purchase intention.

By contrast, for luxury brands – whose perceived value derives from exclusivity, emotional depth, and artisanal quality (Kapferer and Laurent, 2016; Ko *et al.*, 2019) – the standardized and democratizing nature of VTO interfaces may conflict with consumers' expectations of a high-touch, personalized, and sensorially rich brand experience. As noted in prior literature, luxury consumption is often motivated by symbolic and hedonic gratifications such as identity expression, social status signaling, and emotional resonance (Wiedmann *et al.*, 2017; Holmqvist *et al.*, 2020). Nelissen and Meijers (2011) underlined the importance of social status and social interaction in consuming luxury brand-labeled clothes, and these needs of signaling could extend also to the purchase phase of luxury products. While the use of VTO technology generally improves the online shopping experience, its impact may be less pronounced for luxury brands, where the social signal during purchase holds significant importance. Crucially, our findings should not be interpreted as evidence that "luxury consumers reject digital innovation" or that VTO is inherently inappropriate for luxury. Instead, the results indicate a boundary condition: when a VTO experience is perceived as insufficiently aligned with luxury's symbolic and relational expectations, the attitudinal lift generated by e-TAM is weaker. In this light, our findings suggest that VTO may be perceived as misaligned with the luxury brand experience, leading to weaker attitudinal and behavioral responses in this segment. A plausible implication is that luxury implementation may need to emphasize exclusivity cues, curated service elements, or brand-specific storytelling around the tool to preserve the costly-signal meaning that consumers expect from luxury shopping.

The present research thus identifies a critical boundary condition for the effectiveness of immersive retail technologies: the symbolic congruence between the technological experience and brand identity. VTO should not be considered a universally beneficial tool; rather, its effectiveness depends on how well the experiential attributes of the technology align with the

values and expectations consumers associate with the brand. Accordingly, we avoid overgeneralizing these findings as a universal rule about luxury technology adoption. Our evidence instead suggests that the same VTO functionality can generate different outcomes depending on the brand frame and the meaning consumers attach to the purchase experience. However, it is important to specify that our studies capture attitudinal and intentional consumers' responses to VTO technology rather than observed behavior. By integrating technology acceptance models with brand signaling frameworks, our study offers a more holistic understanding of consumer behavior in digitally mediated shopping contexts and highlights the importance of aligning technological innovation with brand positioning strategies.

5.1 Theoretical contributions

This study makes several contributions to the literature on digital consumer behavior and marketing technology. Firstly, it extends the Experiential Technology Acceptance Model by integrating brand type – a symbolic and context-dependent factor – as a moderator. Although e-TAM has been widely applied to understand technology adoption in utilitarian and hedonic settings (Van der Heijden, 2000, 2004; Venkatesh *et al.*, 2016), few studies have explored how brand symbolism conditions these relationships. Our results contribute novelty by demonstrating a theoretically grounded boundary condition for experiential technology acceptance: the same experiential appraisal of VTO (usefulness, ease, enjoyment) translates into attitudes and purchase intentions differently depending on the symbolic frame activated by brand positioning. By demonstrating that the strength of the e-TAM → attitude → purchase intention pathway varies by brand type, this study highlights the need for a more contingent and context-sensitive approach in modeling consumer responses to digital innovations. In other words, we show that technology acceptance in retail is not only “system-driven” but also “meaning-driven,” as brand symbolism shapes how consumers interpret the value of the same digital affordances.

Secondly, this research conceptually bridges e-TAM with Signaling Theory (Spence, 1973) and Costly Signaling Theory (McAndrew, 2021), thus enriching our understanding of symbolic consumption in technology-mediated environments. In luxury contexts, where consumers derive value from exclusivity and prestige, technologies like VTO – despite offering functional benefits – may erode perceived brand distinctiveness if not carefully integrated. The theoretical significance of this integration is that it reframes VTO from a purely functional interface to a signal-bearing touchpoint: VTO can amplify, neutralize, or even dilute the brand's symbolic signals depending on perceived congruence between the technology experience and the brand's identity. This underscores that consumer technology adoption is not only a function of usability and enjoyment but is also mediated by the symbolic compatibility between brand and technology.

Thirdly, the study contributes to external validity by replicating its effects across two product categories (i.e. clothing and footwear) demonstrating that the moderating role of brand symbolism persists across different fashion segments. This reinforces the generalizability of our conceptual model and responds to recent calls in the literature for more ecologically grounded assessments of technology-consumer interaction (Zhang *et al.*, 2019; Chen *et al.*, 2024). More specifically, observing the same moderated mediation in both apparel and footwear strengthens the inference that the boundary condition is not product-idiosyncratic but tied to the brand meaning consumers bring into technology-mediated shopping episodes.

Overall, our findings highlight the importance of considering brand characteristics when examining the adoption of new technologies, as consumer responses may vary significantly depending on whether the brand is perceived as non-luxury or luxury. Our results align with the growing request for a more context-sensitive approach in technology adoption research (Venkatesh *et al.*, 2016). Taken together, the studies advance technology adoption theory by identifying when experiential acceptance mechanisms are likely to be strongest and by

5.2 Managerial implications

The findings of this research offer important managerial insights for brand and retail managers seeking to implement Virtual Try-On (VTO) technologies in a strategically aligned and consumer-responsive manner. Critically, the results indicate that the effectiveness of VTO technology is not universal but contingent upon brand positioning – requiring differentiated implementation strategies for non-luxury *versus* luxury brands. From an innovation management perspective, this implies that VTO should be treated as a portfolio decision (what to roll out, for whom, and how fast), rather than a uniform “plug-and-play” feature.

For non-luxury brands, which are typically associated with functionality, accessibility, and value-for-money, VTO technologies emerge as a powerful lever to enhance both consumer attitudes and willingness to buy. By reducing perceived risk, streamlining decision-making, and providing a playful and engaging product interaction experience (e.g. Patnaik, 2024; Zhang *et al.*, 2019), VTO tools align well with the utilitarian and hedonic expectations of non-luxury brand consumers. Managers in this segment should prioritize the seamless integration of VTO into the digital shopping journey, ensuring user-friendly, intuitive, and responsive interfaces. Furthermore, digital marketing strategies should clearly communicate the practical benefits of VTO – such as improved fit accuracy, time-saving, and ease of comparison – while also emphasizing the experiential features that enrich product exploration and boost consumer involvement. The results suggest significant return potential on investments in VTO for non-luxury apparel and footwear retailers, particularly in e-commerce environments where product tangibility is limited and uncertainty is heightened. Therefore, we suggest to non-luxury house to prioritize frictionless UX, clear communication of practical benefits, and continuous experimentation on fit accuracy to reduce returns and increase confidence. Operationally, non-luxury firms can adopt a broad rollout strategy (high reach) supported by iterative optimization: A/B testing of interface design, continuous calibration of size/fit recommendations, and the use of VTO analytics to identify friction points in the funnel (e.g. drop-off after try-on) and improve conversion.

By contrast, luxury brands face a more complex set of challenges when adopting VTO technology. While immersive tools can increase consumer interaction, their standardized and democratizing nature may dilute core brand associations tied to exclusivity, craftsmanship, and symbolic value. In this context, VTO may not automatically translate into more favorable consumer responses, and a direct replication of the strategies used by non-luxury brands could risk undermining brand equity. Instead, luxury marketers should consider adopting tailored and brand-consistent VTO experiences that reflect the experiential depth and artisanal refinement that luxury consumers expect. This may involve leveraging VTO technology as part of hybrid customer journeys, where digital interactions serve as a prelude to exclusive, high-touch, in-store experiences. For instance, a virtual fitting session could guide consumers through a curated product selection process, which is then completed in-store with personal consultation or bespoke styling services. Such hybrid models preserve the symbolic capital of luxury while allowing for selective digital innovation. Moreover, luxury-focused VTO applications should emphasize storytelling, aesthetic curation, and personalization, incorporating elements such as AI-driven styling recommendations, high-definition textures, or dynamic narratives that showcase craftsmanship and product heritage. Communicative efforts should be equally refined, highlighting how the technology enhances convenience and engagement without compromising authenticity, exclusivity, or emotional resonance. Managers should thus frame VTO technology not merely as a functional enhancement but as a technological extension of the brand’s identity, reinforcing notions of prestige, artistry, and attention to detail. Translating these insights into action, we outline an implementation roadmap for advancing several innovations in VTO and luxury product

strategies and activities. Specifically, we suggest to luxury houses to deploy tiered-access VTO (e.g. invite-only previews) to preserve scarcity; to invest in fidelity-critical competences and capabilities (e.g. material/texture scanning, realistic drape and lighting, and precise fit calibration) to protect craftsmanship perceptions and willingness to pay; to orchestrate hybrid journeys that bind VTO to boutique appointments, alterations, and after-sales services; and to explore authenticated digital artifacts (e.g. digital certificates or limited AR assets) that link try-on traces to provenance and care, reinforcing rarity and improving customer experiences. To connect these recommendations more directly to innovation management decisions, luxury brands may benefit from a selective rollout approach: piloting VTO first with high-fit product lines (e.g. items where visualization reduces uncertainty but does not replace tactile evaluation), deploying it initially to high-value segments (e.g. loyal customers or appointment-based shoppers), and scaling only after monitoring brand-equity indicators (e.g. perceived exclusivity and craftsmanship perceptions) alongside conversion metrics. This “test-and-learn with brand safeguards” logic allows luxury firms to capture the upside of digital innovation while actively managing the risk of symbolic dilution.

Finally, these findings point to the strategic importance of technology-brand alignment. Managers should avoid one-size-fits-all approaches and instead tailor VTO technology implementation strategies based on their brand’s symbolic positioning and the experiential priorities of their target audience. By aligning technological features with brand meaning, firms can maximize consumer receptivity and maintain coherence between digital innovation and brand equity. In practice, managers can translate our key finding (brand-type boundary condition) into a decision framework: (1) diagnose brand-technology congruence (does VTO reinforce the brand promise?), (2) choose the rollout logic (broad vs selective), (3) orchestrate the end-to-end digital journey (standalone e-commerce feature vs hybrid service journey), and (4) define success metrics that include both performance outcomes (conversion, returns) and brand-equity safeguards (exclusivity, craftsmanship, emotional resonance).

5.3 Limitations and future research

While the present study offers important theoretical and practical contributions, several limitations must be acknowledged, which also open promising avenues for future research. First, the sample was composed of Italian consumers, which, while useful for cultural consistency, limits the cross-cultural generalizability of the results. Luxury consumption and technology adoption are both culturally contingent phenomena influenced by values such as materialism, individualism, uncertainty avoidance, and aesthetic norms. Accordingly, future research should replicate the study across different cultural contexts using matched stimuli to test the robustness of the moderated mediation model across cultural segments. In addition, because both studies relied on convenience/snowball recruitment within a single country, future work should incorporate broader and more diverse sampling frames and explicitly control for relevant consumer background variables (e.g. income bands, online shopping frequency, and luxury shopping experience) to improve generalizability, particularly in luxury contexts, where purchasing norms and category meaning vary substantially across cultures.

Second, although the study investigated two fashion-related product categories (clothing and footwear), the scope could be broadened to other domains where VTO technologies are increasingly adopted, such as cosmetics, eyewear, or furniture. These categories differ in terms of product involvement, tangibility, and symbolic value, and may thus elicit distinct consumer responses to VTO. Relatedly, future research should also test additional luxury categories (e.g. leather goods, jewelry, watches), where the purchase experience may be even more identity-laden and authenticity-sensitive, potentially strengthening or altering the observed boundary condition.

Third, the study relied exclusively on self-reported measures of attitudes and behavioral intentions. While such measures are standard in consumer behavior research, they may not always translate into actual purchase behavior. Future studies should incorporate behavioral or

physiological data – such as clickstream analytics, eye-tracking, or biometric indicators of engagement – to triangulate findings and enhance validity. Moreover, robustness would be strengthened by testing alternative outcome variables beyond willingness to buy, such as willingness to pay, choice-based tasks (e.g. preference or trade-off tasks), or simulated shopping baskets, to assess whether the effects generalize to valuation and choice behavior.

Fourth, the study did not account for individual-level psychological traits that may shape the technology adoption process, such as digital literacy, innovativeness, privacy concerns, or consumers' need for touch (Peck and Childers, 2003). These traits are relevant in virtual consumption contexts where sensory surrogacy and digital competence influence the perceived adequacy of virtual experiences. Including such dispositional variables could improve explanatory power and could enable segmentation strategies tailored to different consumer psychographics. Future research could further probe the boundary condition identified here by varying the “luxury-ness” of the VTO execution itself (e.g. levels of personalization, concierge-style guidance, exclusivity cues), and examining whether consumer traits (e.g. status-seeking orientation) strengthen or weaken the observed moderation.

Fifth, future research could explore longitudinal effects by tracking changes in consumer attitudes and behavior as exposure to VTO technology increases over time. This would allow for the examination of habituation effects, potential shifts in perceived congruence with luxury brand identity, and the long-term impact of digital transformation on brand equity.

Sixth, future work should model dual-process routes (e.g. utilitarian/hedonic *versus* symbolic/status) clarifying when VTO effects are driven by instrumental benefits *versus* identity reinforcement, and testing moderated mediation with self-brand congruity, status motives, and social visibility as contingencies. Relatedly, although the present research draws on theoretical arguments concerning authenticity, psychological distance, and symbolic fit to interpret why the effects of VTO-related evaluations may be weaker in luxury contexts, these mechanisms were not directly measured or empirically tested. Therefore, the proposed explanation should be interpreted as theoretically grounded but inferential. Future research should directly assess these psychological processes, for instance by measuring whether VTO experiences reduce perceived authenticity, increase psychological distance from the product or brand, or weaken the symbolic fit between the digital try-on experience and the luxury consumption context. Testing these mechanisms would provide a more fine-grained understanding of why brand type moderates the relationship between VTO evaluations, attitudes, and purchase-related outcomes.

Finally, as the user experience of actual VTO tools – characterized by interactivity, motion tracking, and device responsiveness – is more complex, future studies should incorporate live or interactive VTO platforms (e.g. web-based AR try-ons, mobile-based applications) to better capture the experiential richness of the technology and its impact on emotional engagement and decision-making processes. Importantly, the present studies relied on an imagined VTO scenario supported by visual stimuli; while this approach enables experimental control, it may weaken ecological validity. This means that, because participants evaluated a scenario-based VTO experience rather than directly interacting with an operational VTO tool, the findings should be interpreted as evidence of consumers' responses to VTO adoption cues rather than as behavioral evidence of actual VTO use. Future research should therefore complement scenario-based experiments with real usage designs (e.g. field experiments or lab studies using working VTO tools) to assess whether the observed effects replicate under actual interaction conditions.

(The Appendix follows overleaf)

Table A1. Measurement items, sources, and reliability of study 1 and 2 instruments

Variables	Measurement items	Source (<i>adapted from</i>)	Cronbach alpha	
			Study 1	Study 2
<i>Perceived usefulness of VTO (PU)</i> “1” = strongly disagree; “7” = strongly agree	VTO improves my online shopping productivity VTO enhances my effectiveness when shopping online VTO is helpful in buying what I want online VTO improves my online shopping ability	Kim and Forsythe (2008b)	$\alpha = 0.94$	$\alpha = 0.92$
<i>Perceived ease of use of VTO (PEOU)</i> “1” = strongly disagree; “7” = strongly agree	Using VTO is clear and understandable Using VTO does not require a lot of mental effort VTO is easy to use	Kim and Forsythe (2008b)	$\alpha = 0.91$	$\alpha = 0.92$
<i>Perceived enjoyment of VTO (PE)</i> “1” = strongly disagree; “7” = strongly agree	Shopping with VTO is fun for its own sake Shopping with VTO is exciting Shopping with VTO is enjoyable Shopping with VTO is interesting	Kim and Forsythe (2008b)	$\alpha = 0.95$	$\alpha = 0.94$
<i>Attitude towards the VTO (ATT)</i> “1” = strongly disagree; “7” = strongly agree	VTO would be of good experience VTO would be of superior feel VTO would be of pleasant shopping environment VTO would be interesting and worthwhile	Chidambaram et al. (2024)	$\alpha = 0.87$	$\alpha = 0.86$
<i>Willingness to buy (WTB)</i> “1” = very low; “7” = very high	The likelihood of purchasing this product is The probability that I would consider buying the product is My willingness to buy the product is	Dodds et al. (1991)	$\alpha = 0.92$	$\alpha = 0.93$
<i>Overall Experiential Technology Acceptance Model (overall e-TAM)</i>	Perceived usefulness of VTO (PU) Perceived ease of use of VTO (PEOU) Perceived enjoyment of VTO (PE)	Van der Heijden (2000, 2004)	$\alpha = 0.81$	$\alpha = 0.80$
<i>Personal Innovativeness (INN)</i> “1” = strongly disagree; “7” = strongly agree	When I hear about a new technology, I look for ways to experiment with it Among my peers, I am usually the first to try new technologies I like to experiment with new technologies	Kim and Forsythe (2008a)	$\alpha = 0.88$	$\alpha = 0.86$

Source(s): Authors’ own work

Notes

1. As an additional robustness check, brand familiarity was also assessed. Familiarity differed across brands, with Zara® being more familiar than Gucci® and Valentino®. However, an ANCOVA showed that the effect of brand condition on perceived luxury remained significant after controlling for familiarity, $F(2, 123) = 70.31, p < 0.001$, whereas familiarity itself was not a significant predictor of perceived luxury, $F(1, 123) = 0.43, p = 0.515$. Thus, the observed differences in perceived luxury cannot be attributed to brand familiarity.

2. *Stimulus in non-luxury brand condition:* A well-known non-luxury brand (Zara®) has decided to adopt Virtual Try On (VTO) technology during the online shopping experience for clothing. Imagine being in the comfort of your own home and living this online browsing experience that allows the use of Virtual Try On (VTO) technology. Now, assuming that the product reflects your taste and you have the economic availability to make the purchase, we ask you to express your opinions regarding the following statements ...

Stimulus in luxury brand condition: A well-known luxury brand (Valentino®) has decided to adopt Virtual Try On (VTO) technology during the online shopping experience for clothing. Imagine being in the comfort of your own home and living this online browsing experience that allows the use of Virtual Try On (VTO) technology. Now, assuming that the product reflects your taste and you have the economic availability to make the purchase, we ask you to express your opinions regarding the following statements ...

3. *Stimulus in non-luxury brand condition:* A well-known non-luxury brand (Zara®) has decided to adopt Virtual Try On (VTO) technology during the online shopping experience for footwear. Imagine being in the comfort of your own home and living this online browsing experience that allows the use of Virtual Try On (VTO) technology. Now, assuming that the product reflects your taste and you have the economic availability to make the purchase, we ask you to express your opinions regarding the following statements ...

Stimulus in luxury brand condition: A well-known luxury brand (Gucci®) has decided to adopt Virtual Try On (VTO) technology during the online shopping experience for footwear. Imagine being in the comfort of your own home and living this online browsing experience that allows the use of Virtual Try On (VTO) technology. Now, assuming that the product reflects your taste and you have the economic availability to make the purchase, we ask you to express your opinions regarding the following statements ...

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495

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